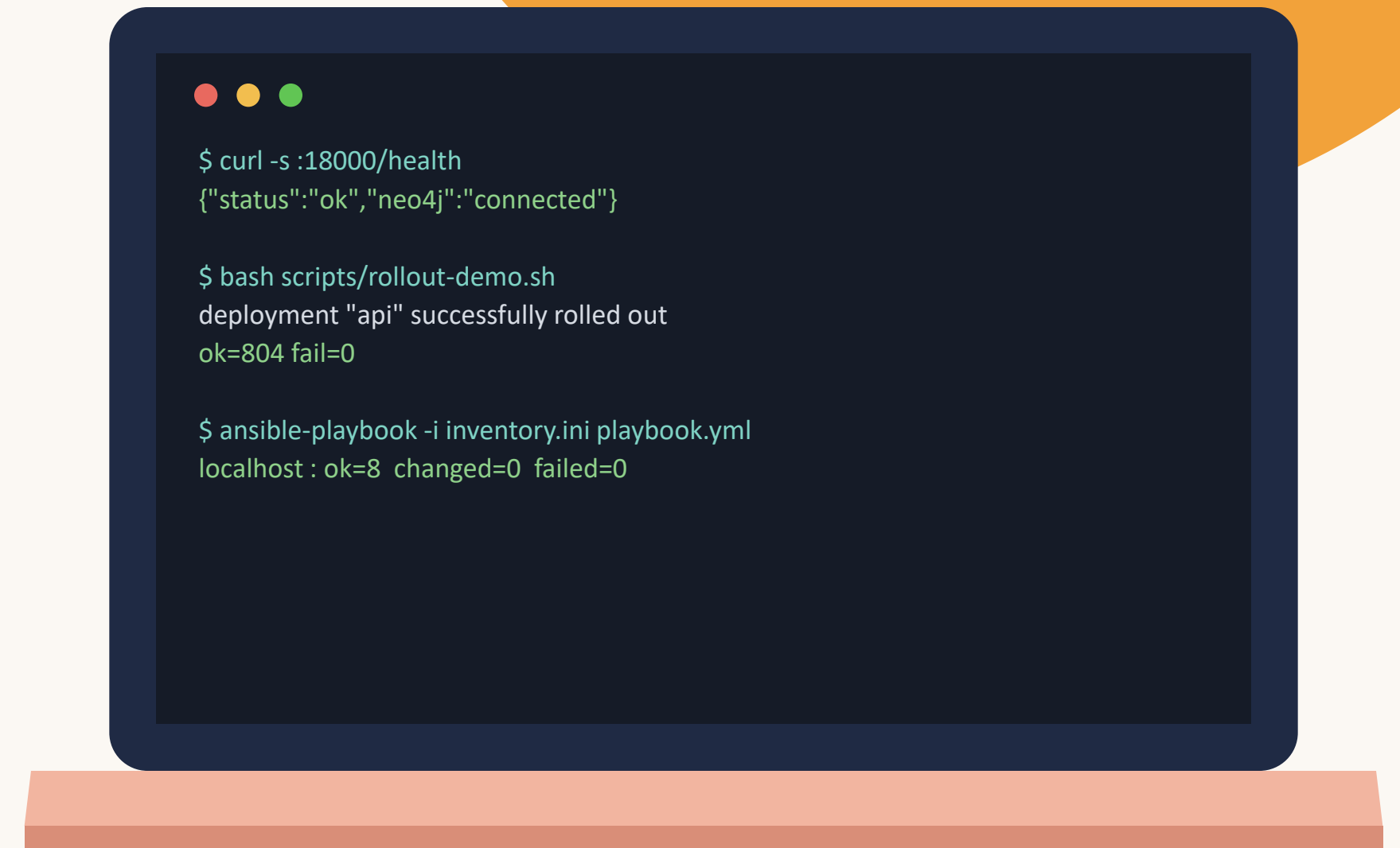


DevOps CA-2 TruthGraph

Reflection & Report: one verification API taken through CI/CD, configuration, release and monitoring

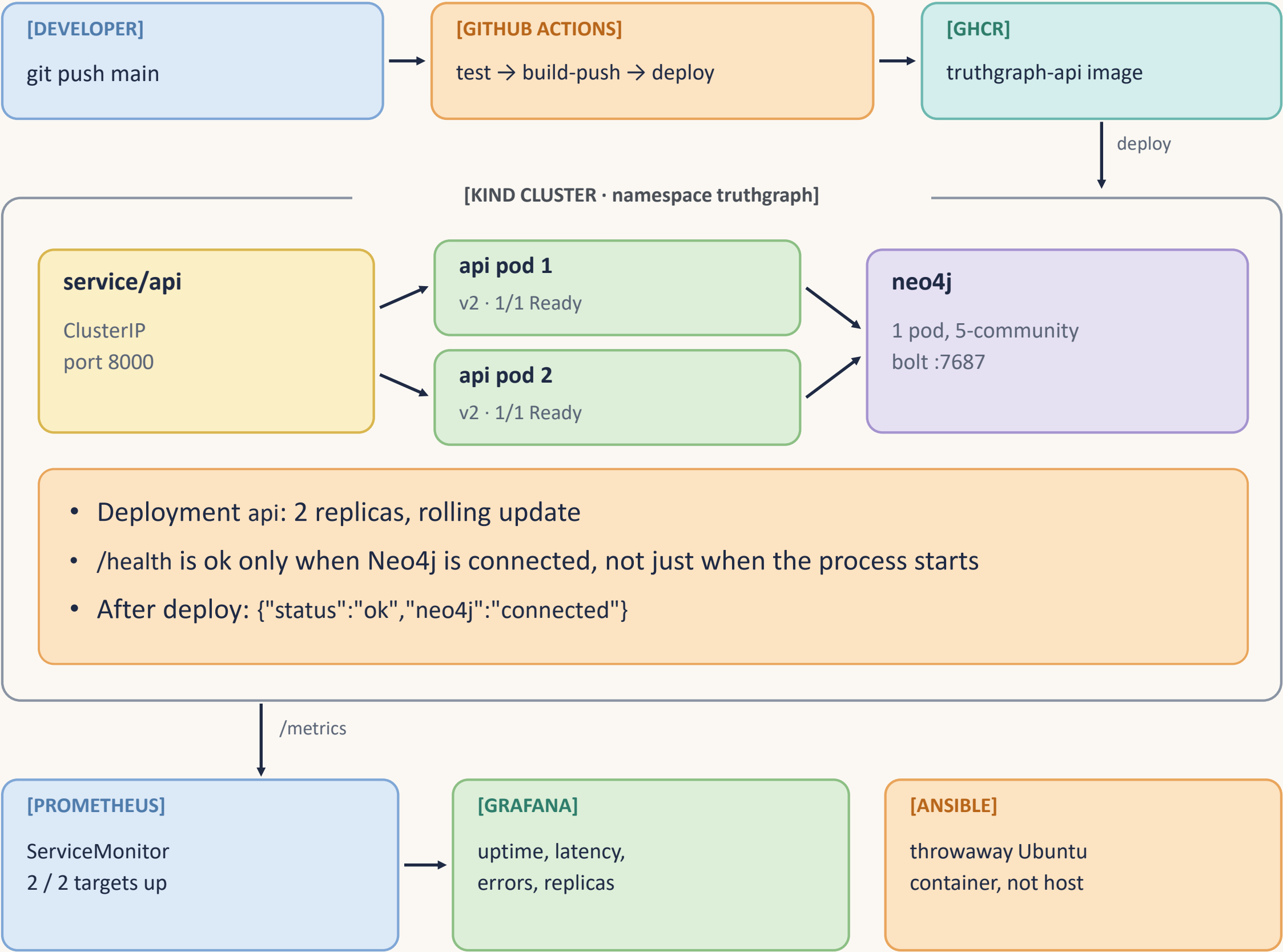
Shubhankar Sarangi

PRN: 23070122206



Architecture

One service, the TruthGraph API, through build, configuration, release and monitoring.



LAYER	WHAT I USED
Service	TruthGraph API, Python 3.11
Graph DB	Neo4j 5 community
Images	Docker buildx, pushed to GHCR
Cluster	kind, kindest/node v1.37.0
CI/CD	GitHub Actions, 3 jobs
Config	Ansible playbook
Monitoring	Prometheus + Grafana

PODS AND SERVICES IN THE truthgraph NAMESPACE

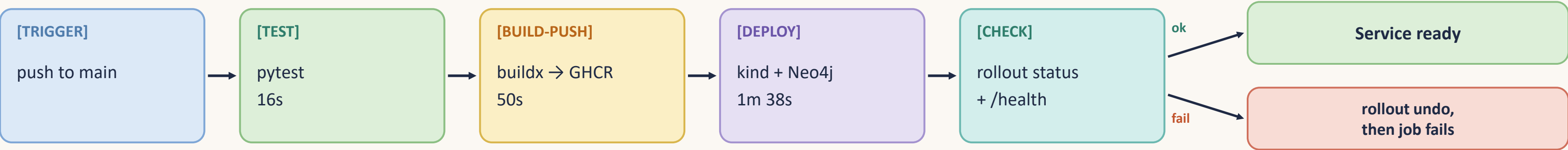
```
kubectl --namespace truthgraph get pods,svc
NAME                                READY   STATUS    RESTARTS   AGE
pod/api-dc785d97bd-82xkp            1/1     Running   0           12m
pod/api-6c785d97bd-n62d2            1/1     Running   0           12m
pod/neo4j-7cbfb4b68b-7rx26          1/1     Running   0           12m

NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
service/api ClusterIP    10.96.233.137 <none>        8000/TCP         16m
service/neo4j ClusterIP    10.96.63.64   <none>        7687/TCP,7474/TCP 16m

tripa@SideChick:~/shubh$ kubectl --namespace truthgraph exec deploy/api -- python -c '... /version and /health'
{"service":"truthgraph","version":"v2"}
{"status":"ok","neo4j":"connected"}
```

Pipeline Flow: GitHub Actions

truthgraph-ci-cd.yml runs on a push to main, on pull requests and on manual dispatch. Three jobs, in order.



Job 1: test

- The suite mocks the outside calls, so CI needs no API keys
- Result: **29 passed in 0.36s**

16s

Job 2: build-push

- Builds the image with buildx
- Pushes to GHCR only on a real push; a pull request builds and stops

50s

Job 3: deploy

- Loads the image and Neo4j into kind, applies the manifests, calls /health
- If the check fails, rolls the API back and fails the job

1m 38s

LATEST RUN #2: SUCCESS IN 2M 58S

truthgraph-ci-cd

✓ Add the TruthGraph DevOps CA-II pipeline so the service can be tested... #2

Summary

All jobs

test

build-push

deploy

Run details

Usage

Workflow file

Triggered via push 8 minutes ago

blacroc10 pushed -> 3075004 main

Status

Success

Total duration

2m 58s

Artifacts

1

truthgraph-ci-cd.yml

on: push

test 16s

build-push 50s

deploy 1m 38s

BOTH RUNS GREEN: 2M 58S AND 2M 57S

2 workflow runs

Workflow

Event

Status

Branch

Actor

✓ Add the TruthGraph DevOps CA-II pipeline so the service can be tested... main 8 minutes ago 2m 58s

truthgraph-ci-cd #2: Commit 3075004 pushed by blacroc10

✓ Add the TruthGraph DevOps CA-II pipeline so the service can be tested... main 11 minutes ago 2m 57s

truthgraph-ci-cd #1: Commit 9202d4f pushed by blacroc10

Deploy is more than half of each run. Tests and the image build together take about a minute.

Challenges Faced During Implementation

TESTS CALLED OUTSIDE SERVICES

- **Problem:** Running the real calls in CI would need API keys.
- **Solution:** Mocked the outside calls. **29 passed** in **0.36s** with no keys.

PULL REQUESTS COULD PUBLISH IMAGES

- **Problem:** Every run would push an image to GHCR, even from an unmerged PR.
- **Solution:** Push only on a real push to main. A PR builds the image and stops.

"STARTED" IS NOT "HEALTHY"

- **Problem:** The API process can be up while Neo4j is unreachable.
- **Solution:** /health returns ok only when Neo4j is connected: "neo4j":"connected".

CONFIGURING WITHOUT TOUCHING THE HOST

- **Problem:** A playbook run against the real machine is hard to undo.
- **Solution:** Aimed Ansible at a throwaway container: user truthsvc, app.conf, app files.

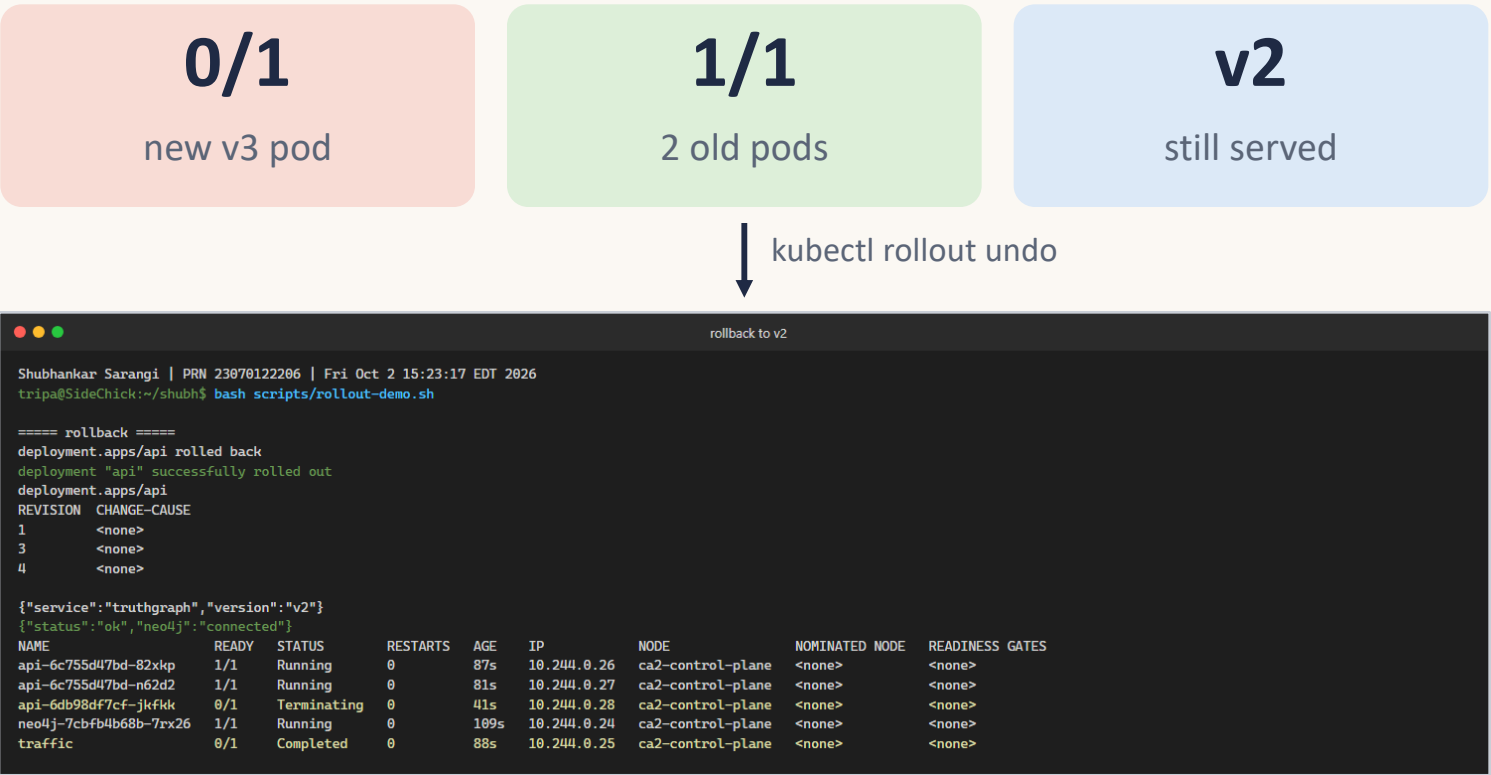
[A BAD RELEASE MUST NOT TAKE THE API DOWN]

v3 was built so new pods fail readiness. The rollout timed out (rollout_status_exit=1).

```
failed release, readiness stays 0/1

Shubhankar Sarangi | PRN 23878122286 | Fri Oct 2 15:23:17 EDT 2026
tripa@SideChick:~/shubh$ bash scripts/rollout-demo.sh

===== bad v3 (readiness fails) =====
deployment.apps/api env updated
Waiting for deployment "api" rollout to finish: 1 out of 2 new replicas have been updated...
rollout_status_exit=1
NAME          READY   STATUS    RESTARTS   AGE   IP            NODE          NOMINATED NODE   READINESS GATES
api-6c755d47bd-82xkp  1/1     Running   0          86s   10.244.0.26   ca2-control-plane  <none>          <none>
api-6c755d47bd-n62d2  1/1     Running   0          80s   10.244.0.27   ca2-control-plane  <none>          <none>
api-6db98df7cf-jkfkx  0/1     Running   0          40s   10.244.0.28   ca2-control-plane  <none>          <none>
===== service still answers on the previous pods =====
{"service":"truthgraph","version":"v2"}
```



Rollback restored v2, /health was ok again, and the bad pod went to **Terminating**.

Lessons Learned

Healthy means the dependency works

A check that only proves the process is up would have passed with Neo4j down.

Zero downtime is a number

Traffic on /health during v1 → v2: **ok=804, fail=0**. Then /version returned v2.

Readiness blocks, rollback restores

The probe kept v3 out of traffic. It did not undo the release; rollback is its own step.

Monitoring needs real signals

/error returns 500 on purpose, so the **8.49%** error ratio is test traffic, not an outage.

Idempotency is the second run

First apply **changed=8**. Second apply **changed=0**: truthsvc uid 999, group gid 1001.

Fast CI needs no secrets

Mocked tests run in 16s of a 2m 58s pipeline. Deploy (1m 38s) is the slow part.

GRAFANA, LAST 15 MINUTES



PROMETHEUS: BOTH API PODS UP (2 / 2)

serviceMonitor/truthgraph/api/0				Show config	2 / 2 up	
Endpoint	Labels	Last scrape	State			
http://10.244.0.27:8000/metrics	container="api" endpoint="http" instance="10.244.0.27:8000" job="api" namespace="truthgraph" pod="api-6c755d47bd-n62d2" service="api"	6.465s ago 1ms	UP			
http://10.244.0.26:8000/metrics	container="api" endpoint="http" instance="10.244.0.26:8000" job="api" namespace="truthgraph" pod="api-6c755d47bd-82xkp" service="api"	13.371s ago 1ms	UP			

ANSIBLE SECOND APPLY: CHANGED=0

```
ansible-playbook --second apply, changed=0

Shubhankar Sarangi | PRN 23878122286 | Fri Oct 2 15:23:17 EDT 2026
tripa@SideChick:~/shubh$ docker exec -w /work/ansible ca2-ansible-truthgraph ansible-playbook -i inventory.ini playbook.yml
PLAY RECAP *****
localhost          : ok=8   changed=0   unreachable=0   failed=0   skipped=0   rescued=0   ignored=0

tripa@SideChick:~/shubh$ docker exec ca2-ansible-truthgraph id truthsvc
uid=999(truthsvc) gid=1001(truthgraph) groups=1001(truthgraph)

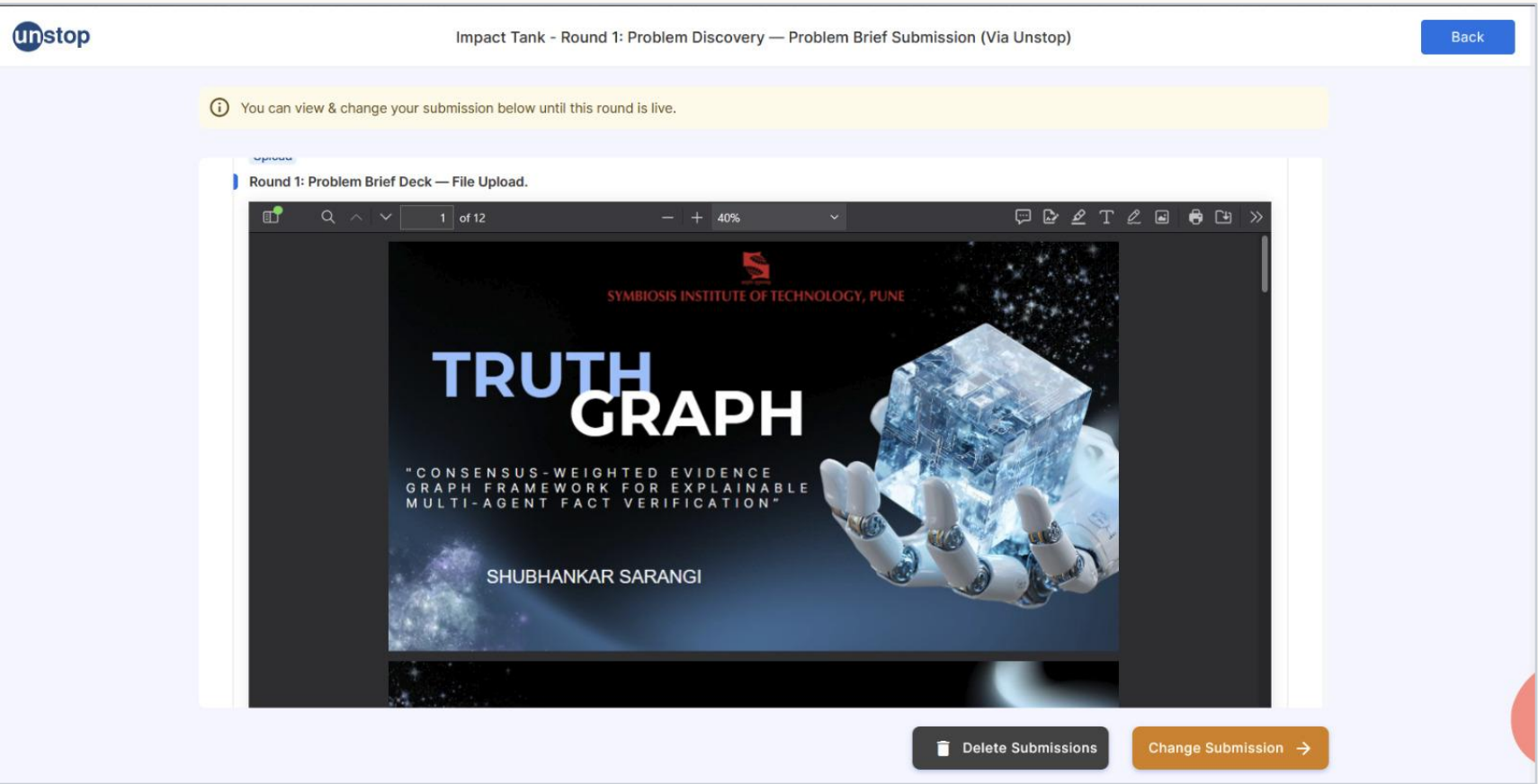
tripa@SideChick:~/shubh$ docker exec ca2-ansible-truthgraph cat /etc/truthgraph/api/app.conf
# Managed by Ansible. Do not edit on the host.
service_name=truthgraph
app_version=v1
log_level=info
app_dir=/opt/truthgraph/api
listen_port=8000
neo4j_uri=bolt://neo4j:7687
```

p99 latency briefly reached about 60 ms during the load-script spikes; p50 and p95 stayed low.

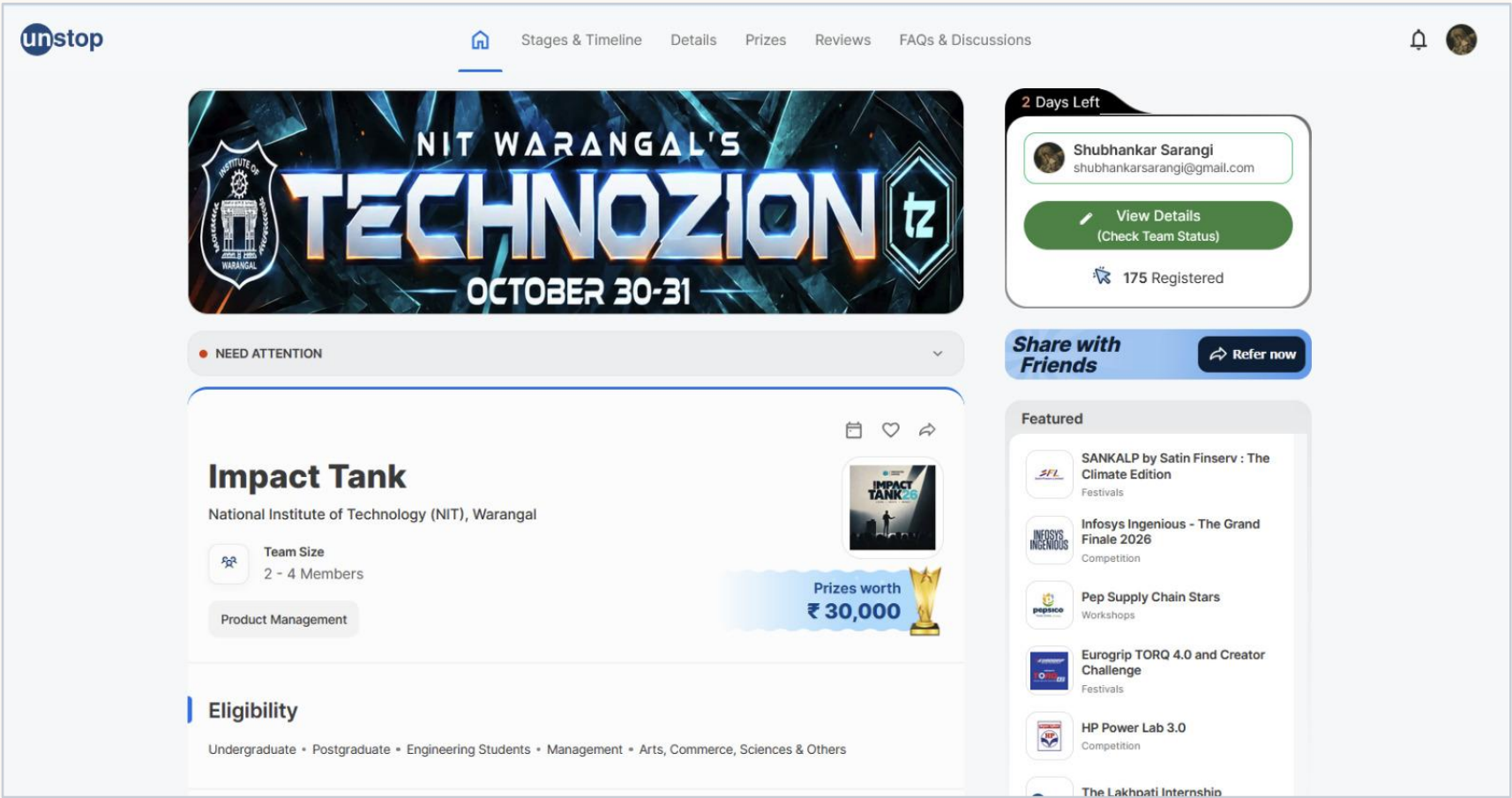
Hackathon Submission: Impact Tank

TruthGraph was submitted to Impact Tank at Technozion '26, NIT Warangal, through Unstop.

ROUND 1 SUBMISSION: TRUTHGRAPH PROBLEM BRIEF DECK



REGISTERED FOR IMPACT TANK, NIT WARANGAL



ROUND
Round 1: Problem Discovery

SUBMITTED
Problem Brief Deck, 12 pages

EVENT
Technozion, October 30 to 31

TEAM SIZE
2 to 4 members

The deck title: "TruthGraph: Consensus-Weighted Evidence Graph Framework for Explainable Multi-Agent Fact Verification".